Vision

PREPARED BY Energy Hippo implementation team

Hippo Synthetic Customer is a controlled fake customer that can be loaded into dev and demo environments to prove that ingestion, reports, gateways, and integrations still work end to end.

The project should behave like a small internal product:

- It has named releases of synthetic data.
- It has a source model that explains why the data exists.
- It generates artifacts instead of storing disconnected copies.
- It includes validation checks for expected downstream behavior.

Why This Exists

Hippo has many integrations and data paths. Testing them independently misses failures that only appear when customer data moves across systems. A synthetic customer gives us one stable reference point that can be loaded repeatedly and used for demos, regression testing, and gateway validation.

Core Design

The repository owns a synthetic source database and scenario manifests. Generator code will transform that source into the same kinds of artifacts Hippo already ingests:

- MDEF files.
- BIF files.
- Backfill inputs.
- Utility-provider-like data files.
- Gateway payloads.
- Integration export payloads.

Those outputs should be reproducible. If two developers generate `DemoUniversityV1` from the same commit, they should get the same customer, accounts, meters, bills, readings, and expected assertions.

Data Safety

The customer identity must be fictional. Real public context is allowed where it is useful and safe:

- Public weather stations.
- Public utility territory assumptions.
- Public geography.
- Fictional addresses in real regions.

Private customer data must not be copied into this repository, even if anonymized.

Rollout Plan

Northlake is built once, through the EEM applications, in ten stages plus one for the services outside the database: the foundation (companies, providers, calendars, security), the hierarchy, the gateways and their first data, rates, bills, the financial interface, tenant allocation, sustainability, operations, and finally the backup that becomes the reference database, and then the identity provider, broker, mail and other services a second server needs beside it. The roadmap says what each stage unlocks and the implementation guide says how to work one. When a capability is added to EEM, Northlake gets the data that exercises it, in the stage it belongs to.

Quality Bar

A scenario is not complete until it has:

- Source data.
- Generated ingestion artifacts.
- A documented reason for existing.
- Expected outcomes.
- A repeatable validation path.